

Lab 7: Molecular Genetics — From DNA to Protein

BIOL-1

Name: _____ **Date:** _____

Overview

The "Central Dogma" of molecular biology describes how the information in your genes becomes a working part of your body: **DNA** → **RNA** → **Protein**. Every protein in every cell — from the keratin in your hair to the hemoglobin carrying oxygen in your blood — was built by reading a gene through this pathway.

In this lab, you will act as the cell's molecular machinery. You will **transcribe** a gene (copy DNA into mRNA) and then **translate** that mRNA into a chain of amino acids using the genetic code. You will also see firsthand how **mutations** — even small changes in the DNA — can alter the protein product.

Learning Objectives

By the end of this lab, you will be able to:

1. Transcribe a DNA sequence into mRNA using base pairing rules.
2. Translate an mRNA sequence into amino acids using a codon table.
3. Identify the effects of point mutations and frameshift mutations on a protein.
4. Explain why the Central Dogma matters for understanding genetics.

Materials

- This worksheet
 - Codon table (provided in Part 1)
 - Colored pencils (optional)
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Part 1: The Genetic Code Reference

Use this mRNA codon table to decode your mRNA sequences. **How to read it:** Find the 1st base on the left, the 2nd base across the top, and the 3rd base on the right.

1st Base	U	C	A	G	3rd Base
U	Phe (F) / Phe (F) / Leu (L) / Leu (L)	Ser (S) / Ser (S) / Ser (S) / Ser (S)	Tyr (Y) / Tyr (Y) / STOP / STOP	Cys (C) / Cys (C) / STOP / Trp (W)	U / C / A / G
C	Leu (L) / Leu (L) / Leu (L) / Leu (L)	Pro (P) / Pro (P) / Pro (P) / Pro (P)	His (H) / His (H) / Gln (Q) / Gln (Q)	Arg (R) / Arg (R) / Arg (R) / Arg (R)	U / C / A / G
A	Ile (I) / Ile (I) / Ile (I) / Met (M) START	Thr (T) / Thr (T) / Thr (T) / Thr (T)	Asn (N) / Asn (N) / Lys (K) / Lys (K)	Ser (S) / Ser (S) / Arg (R) / Arg (R)	U / C / A / G
G	Val (V) / Val (V) / Val (V) / Val (V)	Ala (A) / Ala (A) / Ala (A) / Ala (A)	Asp (D) / Asp (D) / Glu (E) / Glu (E)	Gly (G) / Gly (G) / Gly (G) / Gly (G)	U / C / A / G

Tip: Each cell shows 4 amino acids separated by slashes, corresponding to 3rd base = U, C, A, G (top to bottom).

Remember: AUG is both the Start codon AND codes for Methionine (Met). UAA, UAG, UGA are Stop codons.

Part 2: Transcription and Translation

Transcription is like copying one page from a library book — the original DNA stays in the nucleus, and the mRNA copy travels to the ribosome.

Your gene (DNA template strand): 3' — T A C · C G T · A C G · T C G · G G T ·
G A C · A T T — 5'

Base pairing rules for transcription: A → U, T → A, G → C, C → G

1. Transcribe: Write the mRNA sequence by replacing each DNA base with its RNA complement. Group into codons (3 letters each).

2. Translate: Use the codon table to convert each mRNA codon into an amino acid. Write the amino acid chain.

3. Which codon did you start with? What amino acid does it code for?

4. What told you to stop translating? What was the stop codon?

Part 3: Mutation Analysis

Mutations are changes in the DNA sequence. Some are harmless; others can be devastating. Let's see why.

Return to the **original DNA template strand** from Part 2.

Scenario A: Point Mutation (Substitution)

The **10th base** (the first T in the third codon ACG) changes to a **G**.

New DNA template: 3' – T A C · C G T · A G G · T C G · G G T · G A C · A T T
– 5'

5. Transcribe and translate the mutated sequence.

• Mutated mRNA:

• Mutated amino acids:

6. Did the amino acid chain change compared to your original? What type of mutation is this — silent, missense, or nonsense?

Scenario B: Frameshift Mutation (Insertion)

Return to the **original DNA template**. An extra **G** is inserted right after the 3rd base (after the C in TAC).

New DNA template: 3' – T A C · G C G · T A C · G T C · G G G · T G A · C A T
· T – 5'

7. Transcribe and translate the mutated sequence.

• Mutated mRNA:

• Mutated amino acids:

8. Compare your frameshift protein to your original protein. How many amino acids changed? Why is this called a "frameshift"?

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9. Which type of mutation is more dangerous — the point mutation (Scenario A) or the frameshift (Scenario B)? Explain why.

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Part 4: Putting It All Together

10. Fill in the Central Dogma:

DNA → ___ → ___

11. Where does transcription happen in a eukaryotic cell?

12. Where does translation happen?

13. What enzyme carries out transcription?

14. What structure carries out translation?

15. In your own words, explain why a change in DNA can change what a person looks like or how their body works. (Hint: trace the path from gene to trait.)